W-MEAF for Advertisement Video QA

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Methodology

Whisper-Guided Multi-Path Evidence and Answer Fusion (W-MEAF) is a test-time reasoning pipeline for advertisement video QA. It uses Qwen2.5-VL-7B-Instruct as the multimodal backbone and faster-whisper base for speech transcription. The final submitted score is **0.6495**. For each sample, OpenCV decodes video frames, faster-whisper transcribes narration, and Qwen2.5-VL constructs a compact evidence brief covering product identity, visual demonstrations, readable text, spoken claims, offer terms, and consumer benefits before generating an answer.

W-MEAF generates three complementary branch answers. **Dense Keyframe Evidence** combines 8 uniformly sampled frames with visually scored keyframes sampled at 1 fps, retaining up to 16 frames at 589,824 pixels per frame. Keyframes are selected by sharpness, edge density, motion, late-video position, temporal separation, and average-hash diversity. **Reranked Candidate Selection** uses the same dense visual-audio evidence, generates three candidate answers with temperature 0.4 and top-p 0.9, and selects one answer using a deterministic judge prompt. **Duration-Adaptive Evidence** samples 6, 8, 10, or 12 lower-resolution frames according to video length, using a 262,144-pixel budget to preserve global temporal context.

Conservative Answer Fusion is a text-only Qwen2.5-VL stage. It receives the three branch answers for each of the 3,108 samples, treats the reranked answer as the primary hypothesis, and adds complementary details only when they are consistent and question-relevant. The key idea is branch-level evidence diversification followed by conservative fusion: dense keyframes capture short informative moments, duration-adaptive sampling covers long-range context, reranking improves answer selection, and Whisper supplies spoken advertisement claims within a 7B-class inference budget.